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Differential responses of low- and high-flow dairy cows to automatic cluster removal and dynamic pulsation settings

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ABSTRACT

The objective of this study was to quantify the effects of automatic cluster removal (ACR) and dynamic pulsator settings on milk yield, milking time, and milk flow characteristics in dairy cows differing in intrinsic milk flow capacity. A total of 104 spring-calving cows were classified as high- or low-flow and assigned to 4 milking treatments combining static or dynamic pulsation with 2 ACR switch-points (0.2 or 0.8 kg/min). Cows rotated through all treatments in a balanced 8-week crossover design. Milking duration, flow variables, and teat condition scores were analyzed using mixed models with repeated measures. Increasing the ACR threshold from 0.2 to 0.8 kg/min reduced total milking duration by 25% to 28% and overmilking time by 27 to 28 s without affecting yield or milk composition. Dynamic pulsation exerted smaller but consistent effects, slightly increasing average milk flow rate (+1%). Significant treatment × flow rate interactions were detected for milking duration and average milk flow rate, indicating greater relative efficiency gains among low-flow cows. The probability of firm teat ends declined from 8.4% to 3.8% and ringed teat barrels from 8.3% to 4.8% as the ACR threshold increased. Teat tissue responses were similar across flow rate classes, whereas efficiency traits remained flow dependent. These findings provide both physiological and practical evidence that adaptive detachment and pulsation control can improve milking efficiency and teat health across heterogeneous cow populations.
Key words: milking, cluster removers, dynamic pulsation, milking efficiency

INTRODUCTION

Automation and precision management have driven major advances in modern milking technology over the past 2 decades. The objectives of milking system design have evolved from achieving basic throughput to optimizing the physiological and mechanical interaction between the cow and the machine. Cluster detachment strategies, pulsator control, and vacuum management now aim to minimize overmilking, protect teat tissue, and improve overall milking efficiency. Automatic cluster removal (**ACR**) and dynamic pulsation systems are central to this evolution, yet their performance depends on both machine configuration and cow-level milk flow characteristics (Rasmussen, 1993; Mein et al., 2001; Edwards et al., 2013; Upton et al., 2025a).

Automatic cluster removal systems were originally introduced to prevent overmilking and reduce teat end congestion by terminating milking when milk flow declines below a set threshold. Studies have shown that raising the ACR switch-point from 0.2 to 0.8 kg/min can substantially reduce milking duration in pasture-based systems without compromising yield, strip milk yield, or SCC (Edwards et al., 2013; Upton et al., 2023). These findings indicate that modern systems can safely operate at higher ACR thresholds to improve parlor efficiency, although the extent of benefit varies among cows and herds. Results of a similar nature have been confirmed in higher-yielding herds on 3 times daily milking and at even higher ACR switch-point settings (Wieland et al., 2020).

Dynamic pulsation systems have emerged as a complementary approach to attempt to increase milking speed. These systems adjust pulsator rate and ratio within individual milking events according to milk flow, with the aim of increasing the effective b-phase during high flow and extending the d-phase during low flow to enhance comfort and circulation (Upton et al., 2025a). Although earlier work combined dynamic vacuum and pulsation control based on milk flow rate (Ambord and Bruckmaier, 2010), most subsequent research has focused on flow-responsive vacuum adjustment rather than pulsation phase control (Stauffer et al., 2020; Reinemann et al., 2021). Ströbel et al. (2013) modeled liner wall movement and vacuum behavior during the b- and d-phases to optimize control strategies at different flow rates. Some recent studies have evaluated dynamic or flow-responsive pulsation systems designed to vary pulsation

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rate and ratio within individual milkings to improve teat condition and milking performance (Singh et al., 2024). Collectively, these findings indicate that dynamic pulsation can align the mechanical action of the milking unit with near real-time milk flow, although their combined interactive effect with cow-level milk flow rates have yet to be quantified.

Cows within herds display large variation in milk flow properties, influenced by teat canal anatomy, oxytocin release dynamics, udder storage fraction, and milking preparation (Bruckmaier et al., 1995; Tančin et al., 2006). High-flow cows typically exhibit shorter milking times and higher peak flow rates, whereas low-flow cows are more susceptible to prolonged milking durations and increased vacuum at the teat end, especially where the milk line is above the cow standing level. Such cow-level variation challenges the use of uniform machine settings across all animals. Previous research on these topics has often balanced treatments by average herd flow rate, implicitly assuming that cows respond similarly to a given ACR threshold or pulsation pattern. For example, previous work by this author demonstrated that raising the ACR threshold and applying dynamic pulsation both reduced milking duration and increased average milk flow without compromising yield (Upton et al., 2025a). As a result, although aggregate effects of machine settings on milking performance are well documented, the specific differences in response between low- and high-flow cows have not been explicitly quantified. The present study extends that research by examining whether cows with inherently high or low milk flow characteristics respond differently to these machine settings.

This lack of quantification limits the development of precision milking strategies capable of tailoring settings to individual cows. In the future, increased sensorization of milking equipment will use within milking flow data to adjust control parameters individually (Upton et al., 2025b). To enable this, empirical data are needed to describe how cows of differing intrinsic milk flow properties respond to the combination of ACR threshold and pulsator settings in terms of milking time, yield, and milk flow profile. Understanding these interactions would provide the physiological basis for adaptive control systems that balance efficiency with animal comfort and udder health.

Therefore, the objective of this study was to quantify the effect of ACR and dynamic pulsator settings on milk yield, milking time, and milk flow characteristics in dairy cows classified as low- or high-flow-rate animals. This information will help define performance-based guidelines for the application of dynamic pulsation and ACR thresholds across heterogeneous cow populations in modern milking systems.

MATERIALS AND METHODS

The study was conducted at the Teagasc Animal and Grassland Research and Innovation Centre, Moorepark, Ireland, with ethical approval from the Teagasc Animal Ethics Committee (TAEC0923-393). All procedures complied with the Animal Health and Welfare Act 2013 (updated to September 2023; Law Reform Commission, 2023) and European Directive 86/609/EEC (European Union, 1986).

Cows were milked twice daily in a 46-unit Dairymaster rotary parlor (Dairymaster, Ireland) equipped with simultaneous (4 × 0) pulsation, ACR, and Weighall milk meters. The standard ACR switch-point on the farm was 0.2 kg/min with a 3-s delay. Milking clusters weighed 2.0 kg and were fitted with 1022SF liners (Dairymaster).

The milking software (DairyVue 360, Dairymaster, Ireland) allowed specific pulsator and ACR switch-point combinations to be assigned to individual cows irrespective of milking order. It also enabled dynamic pulsation, automatically switching between low- and high-flow settings once the defined milk flow threshold was reached. Milk flow profiles were recorded for each cow at 5-s intervals.

Cows were managed in a spring-calving, pasture-based system and milked at approximately 16:8 h intervals.

Cow Selection

Cows were eligible for inclusion if their milk SCC was below 200,000 cells/mL at 2 consecutive weekly recordings before the start of the study. From the eligible pool, 104 cows were selected at random. Of these, 84 were Holstein Friesian and 20 were Jersey × Holstein Friesian. At the beginning of the study, cows averaged 218 ± 20 DIM and parity of 2.6 ± 1.8 (Table 1).

Cows were blocked by breed (Holstein Friesian vs. other) and parity (primiparous vs. multiparous) before being randomly assigned to one of 4 treatment groups (n = 26 per group). Each group was made up of 13 cows classified as low-flow (average milk flow rate [**AMF**] = 1.0 kg/min) and 13 as high-flow (AMF = 1.8 kg/min). The cow was considered the experimental unit.

Sample size was determined as part of the Teagasc Animal Ethics Committee approval process (TAEC0923-393), with group size chosen to detect a 5% difference in daily milk yield at $\alpha = 0.05$ and power = 0.80.

Experimental Treatments

Four milking treatments were applied, combining 2 milk flow switch-points for ACR with either static or dynamic pulsator settings:

1. **S0.2**: static pulsation; cluster removed at 0.2 kg/min
2. **D0.2**: dynamic pulsation; cluster removed at 0.2 kg/min
3. **S0.8**: static pulsation; cluster removed at 0.8 kg/min
4. **D0.8**: dynamic pulsation; cluster removed at 0.8 kg/min

Under the static setting, the pulsator ratio was 65:35 with a:b:c:d phase durations of 144, 506, 104, and 246 ms, respectively, across all milk flow rates. For the dynamic setting, the ratio adjusted according to flow rate: when milk flow was below 1.5 kg/min (early and late in milking), the ratio was 63:37 (158, 470, 104, 268 ms), and above 1.5 kg/min, it shifted to 73:27 (150, 577, 110, 164 ms). These were measured using VaDia devices (Biocontrol, Norway).

Pulsation rate was constant at 60 cycles/min, and the system vacuum was maintained at 48 kPa. Before cluster attachment, cows received a premilking routine that included prespraying and drying with paper towels.

Experimental Design

The study commenced on September 21, 2023, and consisted of 4 groups of cows that rotated through the 4 treatments across consecutive 2-wk periods, giving a total duration of 8 wk. At the beginning of the study, each group was randomly assigned one of the 4 treatments.

Teat condition was evaluated by a trained observer after milking on 8 occasions during the study. Teats were scored for congestion of the teat end (**TE**; normal or firm) and ringing of the teat barrel (**TB**; normal or ringed) according to the method of Mein et al. (2001).

A composite milk sample was collected weekly from each cow using the on-farm milk-metering system (Dairymaster Weighall) for analysis of milk composition and SCC using a Fossomatic instrument (Foss).

Strip milk yield was recorded on 4 occasions by reattaching the milking cluster immediately after automatic detachment, adding 2.5 kg of extra weight, and removing the unit once milk flow had visually ceased. Due to the additional time required to complete this procedure during routine milking, strip yield measurements were obtained from 79 cows distributed across all 4 treatments

Data Processing

Standard milking data, including morning and evening milk yields (kg), milking duration (from cluster attachment to automatic removal, s), and AMF, were exported from the farm management software into spreadsheets for analysis. A one-day washout was included between treatment periods.

Additional variables were derived from the 5-s milk flow profiles recorded by the milking system. These included peak milk flow rate (**PMF**; kg/min), dead time (interval from cluster attachment until flow exceeded 0.2 kg/min), overmilking time (duration with flow ≤0.2 kg/min before detachment), and high-flow time (duration with flow >1.5 kg/min). All parameters were determined separately for morning and evening milkings.

Somatic cell count data were $\log_{10}$ transformed before statistical analysis. Data from milkings shorter than 183 s (the minimum cluster-on time for ACR activation) or with total yield <1 kg were excluded, representing less than 1% of all observations. Weekly milk composition results were used to calculate total milk solids (fat + protein, kg) by treatment. Strip yield data were tested for normality using the UNIVARIATE procedure in SAS 9.4 (SAS Institute Inc., Cary, NC).

Statistical Analysis

Milking Data. All data were analyzed using SAS version 9.4. The effects of treatment and flow rate class on the dependent variables were evaluated using the MIXED procedure with restricted maximum likelihood estimation according to the following model:

Table 1. Descriptive statistics of the cows enrolled at the beginning of the study

Item	Mean low[1]	SD	Mean[2]	SD	Mean high[3]	SD
Daily milk yield (kg)	14.2	3.7	16.4	3.9	18.5	3.7
Daily milk duration (min)	14.0	2.6	12.2	2.9	10.4	2.0
Average milk flow rate (kg/min)	1.0	0.1	1.4	0.4	1.8	0.2
DIM	217	22	218	20	218	18
Parity	2.0	2.0	2.6	1.8	3.1	1.9
SCC (×1,000 cells/mL)	38.8	22.5	41.6	30.4	44.5	36.8

[1]Mean of low-flow-rate cows (n = 52).

[2]Mean of all cows (n = 104).

[3]Mean of high-flow-rate cows (n = 52).

$$y = \text{treatment} + \text{flow rate class} + \text{treatment} \times \text{flow rate class} + \text{period} + \text{parity class} + \text{breed} + \text{random (cow)} + \text{repeated (visit)},$$

where y represented milking duration, milk yield, total milk solids per day, AMF, PMF, $\log_{10}$ transformed SCC, dead time, overmilking time, and high-flow time. Treatment included 4 levels (S0.2, D0.2, S0.8, D0.8). Flow rate class referred to high- or low-flow cows based on AMF (1.8 vs. 1.0 kg/min). Period corresponded to the 2-wk experimental phase (1–4), and parity class distinguished primiparous (1) from multiparous (2) cows. Visit distinguished different milking events. Strip yield was analyzed separately using a similar model structure without repeated measures.

Cow was specified as a random effect, with visit treated as the repeated measure within cow. An autoregressive covariance structure (AR(1)) was used to account for correlations among repeated observations, assuming homogeneous variances and exponentially declining correlations with increasing time interval.

A second set of models was run with treatment partitioned into its 2 factorial components: milk flow rate switch-point (0.2 vs. 0.8 kg/min) and pulsation type (static vs. dynamic). The interaction between these factors (switch-point × pulsation type) was tested for each variable. No significant interactions were detected ($P > 0.05$); therefore, results are presented by treatment combination.

Teat Scoring Data. Teat end and TB scores were analyzed using a mixed-model approach in SAS 9.4 (SAS Institute Inc., Cary, NC) with the GLIMMIX procedure and Laplace estimation. The statistical model was as follows:

$$y = \text{treatment} + \text{flow rate class} + \text{treatment} \times \text{flow rate class} + \text{parity class} + \text{breed} + \text{period} + \text{visit} + \text{quarter position} + \text{random (cow ID)},$$

where y represented TE (normal or firm) or TB (normal or ringed). Treatment included 4 levels: S0.2, D0.2, S0.8, and D0.8. Flow rate class referred to high- or low-flow cows based on AMF (1.8 vs. 1.0 kg/min). Parity class distinguished primiparous (1) from multiparous (2) cows. Breed referred to Holstein Friesian or Jersey-cross animals. Period (1–4) denoted the 2-wk experimental blocks. Visit identified morning or afternoon milkings, and quarter position indicated front or rear quarters. The treatment × flow rate class interaction was tested and removed from the final model when not significant ($P > 0.05$).

All fixed effects (treatment, flow rate, parity class, breed, period, visit, quarter position) and the random subject effect (cow ID), with quarters nested within cow, were declared as class variables. Odds ratio estimates were obtained for model terms using the ILINK and EXP options in SAS, providing probability-scale least squares means and 95% CI.

RESULTS

Effect of Treatment on Milking Duration

As shown in Table 2, milking duration was strongly influenced by treatment ($P < 0.001$), and showed significant treatment × flow rate class interactions (P interaction <0.05). Milking duration for S0.2 averaged 679 s, which was 135 s (25%) longer than S0.8, whereas D0.2 averaged 680 s, which was 149 s (28%) longer than D0.8. We observed a significant difference between static and dynamic pulsation at the 0.8 ACR level, but not at the 0.2 kg/min level. The direction of the effect was similar in both morning and evening milkings; however, statistical significance was detected only in morning milkings.

A significant treatment × flow rate class interaction was detected for milking duration ($P < 0.001$; Figure 1A, Table 2). Increasing the ACR threshold from 0.2 to 0.8 kg/min reduced milking duration within both flow rate classes, with a greater reduction observed in low-flow cows. Milking duration decreased by 121 s (21%) in high-flow cows and by 177 s (23%) in low-flow cows.

Effect of Treatment on AMF

A significant treatment effect was also detected for AMF ($P < 0.001$), together with a strong treatment × flow rate class interaction (P interaction <0.001). The AMF for S0.2 was 0.23 kg/min (17%) lower than for S0.8, and a similar difference (0.26 kg/min; 18%) occurred between D0.2 and D0.8. At the higher switch-point (0.8 kg/min), dynamic pulsation slightly increased AMF relative to static pulsation ($P < 0.05$) for a.m. milkings but not p.m. milkings.

A significant treatment × flow rate class interaction was detected for AMF ($P < 0.001$; Figure 1B). Increasing the ACR threshold from 0.2 to 0.8 kg/min increased AMF within both flow rate classes. The AMF increased by 0.3 kg/min in high-flow cows (from 1.5 to 1.8 kg/min) and by 0.2 kg/min in low-flow cows (from 0.9 to 1.1 kg/min).

Effect of Treatment on Overmilking

Overmilking time was affected by treatment ($P < 0.0001$). Cows milked under S0.2 spent 28 s longer in overmilking than under S0.8, whereas those under D0.2 spent 27 s longer than under D0.8.

Although the treatment × flow rate class interaction for overmilking time was significant ($P = 0.03$), the ab-

Table 2. Least squares means for main milking outcomes by treatment; *P* interaction shows the significance of the treatment × flow rate class term (i.e., high or low) in each model

	Treatment[1] (kg/min)					
Item	S0.2	D0.2	S0.8	D0.8	*P*-value	*P* interaction[2]
Milk yield (kg)	12.7	12.76	12.56	12.53	0.4	0.68
AM milk yield (kg)	8.91	8.97	8.8	8.79	0.3	0.11
PM milk yield (kg)	3.79	3.75	3.74	3.73	0.8	0.77
Total solids (kg)	1.32	1.32	1.3	1.3	0.3	0.37
Strip milk (kg)	0.46^{b}	0.59^{b}	0.9^{a}	0.87^{a}	0.02	0.12
Milk duration[3] (s)	679^{c}	680^{c}	544^{b}	530^{a}	<0.0001	<0.0001
AM milk duration (s)	387^{c}	385^{c}	322^{b}	310^{a}	<0.0001	<0.0001
PM milk duration (s)	293^{c}	294^{c}	222^{a}	220^{a}	<0.0001	0.10
PMF[4] (kg/min)	2.78^{b}	2.88^{a}	2.8^{b}	2.9^{a}	0.002	0.74
AM PMF (kg/min)	2.98^{b}	3.11^{a}	3.01^{b}	3.1^{ab}	0.01	0.05
PM PMF (kg/min)	2.57^{b}	2.65^{ab}	2.59^{b}	2.7^{a}	0.05	0.54
AMF[5] (kg/min)	1.18^{a}	1.18^{a}	1.42^{b}	1.44^{b}	<0.0001	<0.0001
AM AMF (kg/min)	1.48^{c}	1.48^{c}	1.71^{b}	1.75^{a}	<0.0001	0.00
PM AMF (kg/min)	0.79^{a}	0.79^{a}	1.01^{b}	1.02^{b}	<0.0001	0.03
Dead time[6] (s)	70^{a}	71^{a}	62^{b}	62^{b}	<0.0001	0.29
AM dead time (s)	26^{a}	26^{a}	23^{b}	23^{b}	0.0001	0.78
PM dead time (s)	45^{a}	45^{a}	40^{b}	40^{b}	<0.0001	0.01
Overmilking time[7] (s)	30^{a}	30^{a}	2^{c}	3^{b}	<0.0001	0.03
AM Overmilking time (s)	15^{a}	15^{a}	0^{b}	1^{b}	<0.0001	0.33
PM Overmilking time (s)	15^{a}	16^{a}	1^{b}	2^{b}	<0.0001	0.02
High flow time[8] (s)	186^{b}	189^{b}	196^{a}	197^{a}	0.01	0.01
AM high flow time (s)	138^{b}	140^{b}	144^{a}	144^{a}	0.10	0.04
PM high flow time (s)	47^{b}	48^{b}	52^{a}	52^{a}	0.004	0.06
Log_{10} SCC[9]	1.70	1.70	1.72	1.71	0.84	0.10
SCC[10] (× 1,000 cells/mL)	68.63	72.28	88.49	74.47	NA[11]	NA

[a–c]Where the main effect is significant, differing letters within rows indicate significance at the *P* = 0.05 level.

[1]S0.2 cluster removed at a milk flow rate of 0.2 kg/min with static pulsator settings; D0.2 cluster removed at 0.2 kg/min with dynamic pulsator settings; S0.8 cluster removed at 0.8 kg/min with static pulsator settings, and D0.8 cluster removed at 0.8 kg/min with dynamic pulsator settings.

[2]*P* interaction shows the significance of the treatment × flow rate class term (i.e., high or low) in each model

[3]Milking duration = time from cups on to cups off.

[4]PMF = peak milk flow rate.

[5]AMF = average milk flow rate.

[6]Dead time = time from cluster attachment to reach a milk flow rate of >0.2 kg/min.

[7]Overmilking time below 0.2 kg/min toward the end of milking.

[8]High flow time = time above 1.5 kg/min.

[9]Log-transformed SCC.

[10]Back transformed from LSM.

[11]NA = not applicable.

solute differences were minimal (<2 s between flow rate groups) and were not considered biologically relevant.

Effect of Treatment on PMF

The PMF differed among treatments ($P = 0.002$) but showed no interaction with flow rate class (*P* interaction = 0.74). The PMF under the dynamic settings was 3% higher ($P < 0.05$) than under the static settings across both ACR settings. Milk yield, total solids yield, and log_{10} SCC were not affected by treatment ($P > 0.05$).

Effect of Treatment on Teat Congestion

The effect of treatment on TE was significant ($P < 0.001$). The probability of firm teat ends ranged from 3.8% for S0.8 to 8.4% for S0.2 (Table 3). Treatments containing static pulsation settings were not different from their dynamic counterparts. Flow rate class also had a significant effect ($P = 0.0004$), with low-flow cows being 3.5 times more likely to have firm teat ends compared with high-flow cows (OR = 3.53; 95% CI: 1.75–7.12). The predicted probability of a firm TE was 9.7% in the low-flow class compared with 3.0% in the high-flow class, indicating greater congestion for slower milking cows (Table 4).

Treatment effects on TB were also significant ($P < 0.001$). The probability of ringed teat barrels ranged from 3.4% for S0.8 to 8.3% for D0.2 (Table 3). Treatments with higher ACR thresholds (0.8 kg/min) were associated with lower TB scores than those with 0.2 kg/min thresholds, whereas flow rate class was not significant

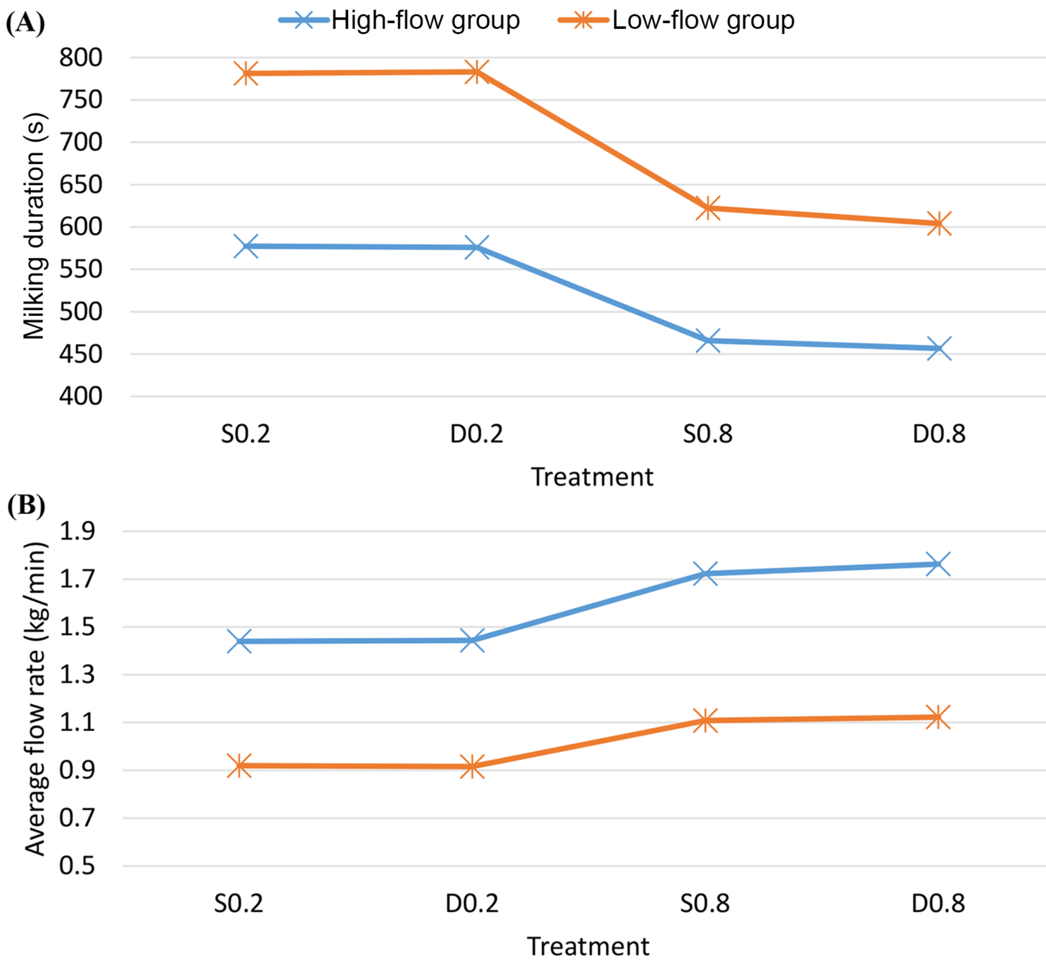

Figure 1. Total milking duration (A) and average milk flow rate (AMF; B) by treatment and flow rate class. Points represent LSM. Treatments were static 0.2 (S0.2), dynamic 0.2 (D0.2), static 0.8 (S0.8), and dynamic 0.8 (D0.8), corresponding to cluster removal at 0.2 or 0.8 kg/min with static or dynamic pulsator settings. Flow rate class was determined from individual cow AMF; the low-flow group averaged 1.0 kg/min, and the high-flow group averaged 1.8 kg/min.

(P = 0.31). Other significant effects in the TB model included period (P = 0.03), time of milking (P = 0.006), and quarter position (P < 0.001), with p.m. milkings and rear quarters showing greater TB congestion. The predicted probability of a ringed TB was 6.6% in the low-flow class compared with 4.6% in the high-flow class (Table 4).

The treatment × flow rate class interaction was tested and found to be not significant in either the TE (P = 0.40) or TB (P = 0.69) models.

Table 3. Predicted probability (P) of firm teat end (TE) and ringed teat barrel (TB)

Treatment[1]	P (firm TE)[2] (%)	95% CI	Tukey group	P (ringed TB)[3] (%)	95% CI	Tukey group
S0.2	8.4	5.2–13.1	a	6.7	4.1–10.8	ab
S0.8	3.8	2.3–6.3	c	3.4	2.0–5.8	c
D0.2	6	3.6–9.6	ab	8.3	5.1–13.1	a
D0.8	4.5	2.7–7.4	bc	4.8	2.9–8.0	bc

[1]S0.2 cluster removed at a milk flow rate of 0.2 kg/min with static pulsator settings; D0.2 cluster removed at 0.2 kg/min with dynamic pulsator settings; S0.8 cluster removed at 0.8 kg/min with static pulsator settings, and D0.8 cluster removed at 0.8 kg/min with dynamic pulsator settings.

[2]TE = percentage of teat ends scored as firm at the quarter level.

[3]TB = percentage of teat barrels scored as ringed at the quarter level.

Table 4. Predicted probability (P) of firm teat end (TE) and ringed teat barrel (TB) by flow rate class

Flow rate class[1]	P (firm TE)[2] (%)	95% CI	P (ringed TB)[3] (%)	95% CI
High-flow	3.0	1.6–5.4	4.6	2.5–8.4
Low-flow	9.7	6.0–15.4	6.6	3.9–11.0

[1]Flow rate class determined from the average milk flow rate of individual cows; low-flow cows averaged 1.0 kg/min, and high-flow cows averaged 1.8 kg/min.

[2]Percentage of teat ends scored as firm at the quarter level.

[3]Percentage of teat barrels scored as ringed at the quarter level.

DISCUSSION

The present study demonstrated that increasing the ACR threshold from 0.2 to 0.8 kg/min markedly reduced total milking duration and overmilking time without compromising yield, milk solids, or SCC. These findings reinforce earlier work in conventional and pasture-based systems showing that truncating the low-flow phase improves milking efficiency while maintaining complete udder emptying (Magliaro and Kensinger, 2005; Clarke et al., 2008; Burke and Jago, 2011; Edwards et al., 2013; Upton et al., 2023, 2025a). The observed 25% to 28% reduction in cups-on time between 0.2 and 0.8 kg/min aligns closely with the 15% to 20% decreases previously reported for similar flow rate thresholds, confirming that excessive extension of the decline phase contributes little to yield but substantially increases exposure of the teat to vacuum and additional liner compression.

Dynamic pulsation exerted smaller (−3% for dynamic settings on 0.8 kg/min ACR settings) but directionally consistent effects. By shortening the liner-closed (d) phase during peak flow and extending it as milk flow declined, dynamic settings slightly increased AMF (+1% for dynamic settings on 0.8 kg/min ACR settings), consistent with its design intent to modulate mechanical load during changing flow conditions. This agrees with previous work showing that modulation of pulsation ratio or phase timing can influence milking dynamics (Bade et al., 2009; Ambord and Bruckmaier, 2010).

The strong relationship between milk flow class and teat end condition highlights the importance of considering animal physiology when interpreting equipment effects. Low-flow cows were 3.5 times more likely to exhibit firm teat ends than high-flow cows, indicating greater tissue congestion under similar machine settings. This supports the concept that slower milk-out prolongs exposure to milking vacuum and liner compression during the decline phase. Similar associations between prolonged low-flow milking and postmilking teat condition have been reported by Hamann et al. (1993), Besier et al. (2016), and Singh et al. (2024). The combination of higher ACR thresholds and dynamic pulsation in the present study therefore limited exposure to adverse mechanical conditions and reduced the predicted probability of firm teat ends from 8.4% to 3.8%. Singh et al. (2024) previously stated that flow-responsive pulsation may foster adequate teat stimulation in cows before the initiation of milk harvest and has the potential to improve teat tissue condition.

Reduced odds of teat barrel ringing at the higher ACR threshold confirm that curtailing overmilking protects teat tissue. Treatments with 0.8 kg/min ACR settings reduced the probability of ringed barrels from 8.3% (D0.2) to 4.8% (D0.8), and from 6.7% (S0.2) to 3.4% (S0.8). The lower incidence of barrel congestion likely contributed to the concurrent reduction in firm teat ends, as both conditions are mechanically linked. Penry et al. (2017) demonstrated that teat barrel and teat end congestion exert additive effects on reducing teat canal cross-sectional area and milk flow rate, indicating that swelling in the barrel can propagate vascular engorgement toward the teat end.

Interestingly, the absence of significant treatment × flow rate class interactions for TE and TB indicates that both high- and low-flow cows responded similarly to the treatments in terms of congestion. This suggests that improved detachment and pulsation control benefit both high- and low-flow cows similarly, reducing mechanical strain by limiting duration of milking under declining flow conditions. In contrast, milking duration and AMF remained highly interactive traits, reflecting that efficiency gains are still flow dependent. The lack of any interaction or main effect on milk yield or total solids yield further demonstrates that yield is not compromised by earlier cluster removal within either flow rate class. Thus, the observed reductions in milking duration and teat congestion arise not from incomplete milk harvest, but from eliminating low-value milking time under high vacuum.

Collectively, these results establish that treatment × flow rate interactions are central to understanding milking efficiency but less relevant to teat tissue response once congestion thresholds are reached. This distinction between process efficiency and teat tissue outcomes provides additional insight into how adaptive detachment and pulsation control may optimize both milking speed and teat health. It also offers a physiological rationale for integrating these strategies, as low-flow cows in this study exhibited greater teat congestion under uniform machine settings than high-flow cows.

CONCLUSIONS

This study demonstrated that increasing the ACR threshold from 0.2 to 0.8 kg/min significantly reduced milking duration and overmilking time without affecting milk yield or composition. Dynamic pulsation exerted smaller but consistent effects on milk flow characteristics. Treatment × milk flow capacity interactions were important for milking efficiency traits, whereas teat tissue responses were similar across high- and low-flow cows. These findings indicate that earlier cluster removal improves milking efficiency without compromising milk harvested, and that cows differing in intrinsic milk flow capacity respond differently in efficiency outcomes but not in teat tissue responses.

NOTES

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Nonstandard abbreviations used: ACR = automatic cluster removal; AMF = average milk flow rate; D 0.8 = dynamic pulsation, cluster removed at 0.8 kg/min; D0.2 = dynamic pulsation, cluster removed at 0.2 kg/min; NA = not applicable; PMF = peak milk flow rate; S0.2 = static pulsation; cluster removed at 0.2 kg/min; S0.8 = static pulsation, cluster removed at 0.8 kg/min; TB = teat barrel; TE = teat end.

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